Miguel Ángel Torres Montoya

Crime Economics · Education Economics · Applied Econometrics · Causal Inference
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EDUCATION

Icesi University, Colombia.
Bachelor of Arts in Economics. GPA: 4.5/5.0. Expected 2027

Graduate-level Coursework:
Quantitative and Computational Methods (4.8/5.0),
Advanced Macroeconomics (4.8/5.0),
Advanced Microeconomics (in progress).
Game Theory and Asymmetric Information (in progress).
Advanced Econometrics (in progress).

Universidad del Rosario, Bogotá, Colombia.
Real Analysis, Bogotá Summer School in Economics (4.7/5.0). [Certificate] 2026

RESEARCH IN PROGRESS

Judges or the System? Causal Evidence on Gender Disparities in Criminal Sentencing in Colombia.
With Eduard F. Martínez and Daniel Mejía.
Undergraduate thesis. Causally identifies the effect of judge sex on sentence length using institutional random assignment (SARJ) across 24 Colombian judicial districts.

Sentencing Process Analysis in the Colombian Accusatory System.
With Eduard F. Martínez-González and Daniel Mejía-Londoño.
Examines the full criminal justice pipeline from criminal notice to final court ruling, identifying procedural bottlenecks and sentencing disparities.

State Capacity and Education Policy Implementation in Colombia.
Staggered difference-in-differences design; examines whether local institutional capacity shapes the effect of the full-school-day reform on dropout

Illicit Drugs in Colombia: The Limits of Eradication and the Tasks Ahead.
With Juan David Gelvez (GIGA) and Eduard F. Martínez.
Examines the cocaine value chain from coca leaf to the European retail market, showing why eradication targets the lowest-value, most easily replaceable link.

RESEARCH EXPERIENCE

2025 – present Research Assistant, Center for Research in Economics and Finance (CIENFI), Universidad Icesi, Cali, Colombia. Supervisor: Eduard F. Martínez-González.

Quarterly Monitoring of Economic Activity in Bogotá with Satellite and Mobility Data.

With Eduard F. Martínez-González.

Cuadernos de Desarrollo Económico No. 83, Secretaría de Desarrollo Económico, Alcaldía Mayor de Bogotá. February 2026

Quarterly GDP estimation at 450m grid level using VIIRS night lights, TransMilenio mobility, and census data; feedforward neural network trained on ~16,000 model configurations with rolling-origin validation. [Report]

Property Registry Revaluation Analysis for Cali. (With the Alcaldía de Cali)

Descriptive analysis of the *reavalúo catastral*, focusing on property owner claims following the reassessment.

2026 – present Research Seedbed, Finance & Investment Club (FIC),

Universidad Icesi, Cali, Colombia.

Contributing researcher in the club's newly founded research division. Designed and delivered a workshop on causal inference methods for undergraduate researchers. [Workshop Slides]

Teaching Experience

2025 – present Teaching Monitor, Universidad Icesi, Cali, Colombia.

Introduction to Business Analytics (2025); Microeconomic Theory III, International Economics, Macroeconomic Theory I (2026).

Big Data and Machine Learning (2026) — graduate elective, M.A. in Economics.

Awards and Honors

2026 1st Place, Undergraduate Presentation Competition on the Behavioral Economics, Universidad Icesi. [Slides]

2025 3rd Place, Undergraduate Presentation Competition on the Economics of Crime, Universidad Icesi. [Slides]

2023 1st Place, Undergraduate Economic Debate Competition, Universidad Icesi.

Additional Training

World Bank Group, via edX.

IEM01x: Evaluating Impact in Low- and Middle-Income Countries. [Certificate] 2026

Technical Skills

Programming: R (Advanced), Python (Intermediate).

Geospatial: Google Earth Engine (rgee), QGIS.

Other Tools: Excel (Advanced), LaTeX, Git, PowerPoint, Canva.

Methods: Machine Learning (XGBoost, k-NN, Neural Networks), Spatial Econometrics, Remote Sensing, Statistical Modeling, Data Visualization, Causal Inference.

Languages

Spanish: Native.

English: Professional Working Proficiency (C1).